

3.4 Paper 2 σ/π and bond energy homework — answers

CIE 9701 AS Chemistry · 8 multiple-choice + 12 written marks

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Section A — Multiple choice

Quick answers

Q	Ans.	Q	Ans.	Q	Ans.
A1	B	A4	C	A7	C
A2	A	A5	D	A8	B
A3	C	A6	B		

Worked explanations

A1. B

2025 ON Paper 12 Q5

In N_2 each nitrogen atom is sp hybridised (one 2s mixes with one 2p). A σ bond is always formed by *head-on* overlap, here $sp-sp$ overlap. Each N uses one sp orbital for the σ bond, one sp orbital for its lone pair, and the two remaining 2p orbitals (at right angles) to form the two π bonds. So the lone pair is in an sp orbital.

A / C: give the lone pair as being in a 2p orbital. After hybridisation the lone pair occupies the unused sp orbital, not a pure 2p orbital.

D: says the σ bond is formed by $s-p$ overlap. In N_2 both atoms are sp hybridised, so the overlap is $sp-sp$.

A2. A

2025 FM Paper 12 Q16

Count the π bonds from the multiple bonds: CO_2 ($\text{O}=\text{C}=\text{O}$) has **two** $\text{C}=\text{O}$ double bonds $\rightarrow 2 \pi$. HCN ($\text{H}-\text{C}\equiv\text{N}$) has one $\text{C}\equiv\text{N}$ triple bond $\rightarrow 2 \pi$. Propene ($\text{CH}_3-\text{CH}=\text{CH}_2$) has only one $\text{C}=\text{C} \rightarrow 1 \pi$. So CO_2 and HCN are the two molecules with two π bonds.

B / C: both include propene, which has only one π bond.

D: omits CO_2 , which also has two π bonds.

A3. C

2022 ON Paper 11 Q5

$\text{H}-\text{C}\equiv\text{C}-\text{C}(\text{CH}_3)=\text{CH}(\text{CH}_3)$ — count every single bond as one σ , and remember that each double/triple bond contributes only **one** σ : $\text{H}-\text{C}$ (1), $\text{C}\equiv\text{C}$ (1 σ), $\text{C}-\text{C}$ (1), the $\text{C}=\text{C}$ (1 σ), $\text{C}-\text{H}$ on the double-bonded CH (1), and the two methyl groups (CH_3 : 1 $\text{C}-\text{C}$ + 3 $\text{C}-\text{H}$ each = 4 + 4). Total = 1+1+1+1+1+4+4 = 13.

A / B: these miss some $\text{C}-\text{H}$ σ bonds (a σ bond exists for every single line and for every $\text{C}-\text{H}$ bond).

D: 16 counts the π bonds as if they were σ bonds — the two π bonds of the alkyne plus the one π bond of the alkene must not be counted.

A4. C

2024 MJ Paper 11 Q26

X is $\text{NC}-\text{CH}=\text{CH}_2$. Count the σ bonds: the $\text{C}-\text{C}$ bond (1), the σ bond inside $\text{C}\equiv\text{N}$ (1), the σ bond inside $\text{C}=\text{C}$ (1), one $\text{C}-\text{H}$ on the CH (1) and the two $\text{C}-\text{H}$ bonds on the CH_2 (2). Total = 6 σ bonds. The $\text{C}\equiv\text{N}$ carbon is sp ; the two $\text{C}=\text{C}$ carbons are sp^2 . There is no CH_3 group, so there are no sp^3 carbons. Row C.

A / B: give a total of 4 σ bonds — the σ bonds inside the $\text{C}\equiv\text{N}$ and $\text{C}=\text{C}$ are omitted.

D: has the correct σ count but wrongly includes sp^3 carbon; X has no saturated (four-single-bond) carbon atom.

A5. D

2025 FM Paper 12 Q39

but-2-ene is $\text{CH}_3-\text{CH}=\text{CH}-\text{CH}_3$. Only the two terminal CH_3 carbons are sp^3 (the two $\text{C}=\text{C}$ carbons are sp^2). Each sp^3 carbon has 4 sp^3 hybrid orbitals, all used for bonding (3 $\text{C}-\text{H}$ and 1 $\text{C}-\text{C}$), so the total is $2 \times 4 = 8$.

A / B / C: these count sp^3 carbon atoms or the orbitals of only one carbon. The question asks for the total number of sp^3 orbitals.

A6. B

2025 ON Paper 11 Q6

NH_4CN is made of NH_4^+ and CN^- ions. NH_4^+ : one of its four $\text{N}-\text{H}$ bonds is a *coordinate* bond (the lone pair on N in NH_3 is donated to H^+); the other three are ordinary covalent bonds. So **1** coordinate bond. CN^- : the $\text{C}\equiv\text{N}$ triple bond contains **2** π bonds. Therefore 1 coordinate bond and 2 π bonds.

A / C: say there are 0 coordinate bonds — but the NH_4^+ ion contains one dative bond.

D: says 3 π bonds; a $\text{C}\equiv\text{N}$ triple bond has only two π bonds (and one σ).

A7. C

2023 ON Paper 12 Q5

In NH_3 the nitrogen is sp^3 hybridised (3 bond pairs + 1 lone pair). When it donates its lone pair to H^+ to form NH_4^+ , nitrogen is still surrounded by four electron pairs and is still sp^3 . The hybridisation therefore does *not* change.

A: NH_4^+ is a regular tetrahedral ion, so its dipole moment is *zero*; NH_3 has a significant dipole moment. The dipole moment decreases, not increases.

B: the $\text{H}-\text{N}-\text{H}$ bond angle *increases* from about 107° (NH_3) to 109.5° (NH_4^+), because the lone pair is replaced by a bonding pair.

D: there is no electron transfer from N to Cl. A *proton* (H^+) is transferred from HCl to NH_3 ; the lone pair on N is *shared*, not transferred.

A8. B

2022 MJ Paper 13 Q9

The average $\text{C}-\text{H}$ bond energy in methane is defined as the enthalpy change when **1 mol of $\text{C}-\text{H}$ bonds** is broken, averaged over the four (non-equivalent) bonds. Since methane has four $\text{C}-\text{H}$ bonds, the equation must break 1 mol of $\text{C}-\text{H}$ bonds in total, i.e. $\frac{1}{4}$ mol of CH_4 to give $\frac{1}{4}$ mol of $\text{C}(\text{g})$ and 1 mol of $\text{H}(\text{g})$ — all species *gaseous* and the $\text{C}-\text{H}$ bonds broken (atoms formed).

A: this is the reverse process (bond *formation*) and would give the negative of the bond energy.

C: this is the *total* atomisation of one mole of methane (breaking *all four* $\text{C}-\text{H}$ bonds), which is $4\times$ the average $\text{C}-\text{H}$ bond energy.

D: this breaks only the *first* $\text{C}-\text{H}$ bond (to give CH_3 and H); it is not an average over all four bonds.

Section B — Mark scheme clips

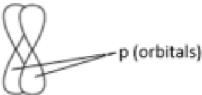
B1(a)(ii) [4]

1(a)(ii)		σ bond	π bond	4
	no of bonds	1	2	
	how orbitals overlap	direct (orbital overlap)	sideways (overlap of adjacent p orbitals)	

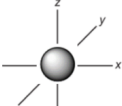
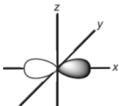
B2(b)(i) [1]

5(b)(i)	sp^2	1
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B3(b)(ii) [2]

5(b)(ii)	M1 σ bond – diagram showing how σ bond is made from direct overlap of orbitals i.e.  OR  M2 π bond - diagram showing how π bond is made from sideways overlap of 2$\times$ p– orbitals i.e.  OR  OR  sideways overlap	2
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B4(c)(iii) [2]

1(c)(iii)	 2s  2p_x	2
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B5(b) [1]

2(b)	CO / it is a polar molecule / it has a (permanent) dipole (but N ₂ is non-polar)	1
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B6(d)(ii) [2]

Question	Answer	Marks
3(d)(ii)	M1 HCl is more stable AND (because) Cl atom is smaller / less distance between electron pair and nucleus / shorter (H-X) bond length M2 (In HCl) stronger attraction between nucleus and shared pair of electrons OR (In HCl) (fewer shells of electrons so) less shielding OR (In HCl) greater orbital overlap so stronger interaction	2